

Joint Rate–Amplitude Modeling for EDA: A Dual-State Marked Point Process

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1 Motivation

Current marked point process (MPP) models for electrodermal activity (EDA) use a single latent arousal state to explain both impulse occurrence and impulse amplitude. In the formulation of Wickramasuriya & Faghih (2020), a latent state x_j evolves as a random walk and drives both the probability of an impulse and its magnitude. While effective, this formulation imposes structural assumptions that are not always physiologically realistic.

1.1 Limitations of Single-State Models

1. **Rate–Amplitude Conditional Independence.**

Given a single latent state, impulse rate and amplitude are conditionally independent, limiting the model’s ability to represent complex physiological coupling.

2. **Inability to Capture Dissociation.**

Empirically, EDA sometimes exhibits dissociations where impulse rate increases while amplitude decreases (or vice versa). A single latent state cannot represent these diverging patterns.

3. **Shared Timescale Constraint.**

Rate and amplitude dynamics are forced to evolve on the same timescale, despite the fact that they may reflect distinct sympathetic mechanisms.

2 Proposed Approach: Dual-State MPP

We propose a **dual-state marked point process** in which impulse rate and amplitude are driven by two interacting latent states:

- $x_j^{(r)}$: a **rate-arousal state** governing impulse occurrence probabilities.
- $x_j^{(a)}$: an **amplitude-arousal state** governing impulse magnitude.

The key idea is to allow these states to be distinct yet coupled, enabling both synchronized and dissociated dynamics.

3 Model Overview

3.1 Latent Dynamics

The latent state evolves as a first-order Vector Autoregressive (VAR) process:

$$x_j = \begin{bmatrix} x_j^{(r)} \\ x_j^{(a)} \end{bmatrix}, \quad x_j = \Phi x_{j-1} + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, Q)$$

The transition matrix

$$\Phi = \begin{bmatrix} \phi_{rr} & \phi_{ra} \\ \phi_{ar} & \phi_{aa} \end{bmatrix}$$

allows each state to influence both itself and the other state.

Interpretation

- **Diagonal terms** (ϕ_{rr}, ϕ_{aa}) capture the persistence or memory of rate and amplitude arousal.
- **Off-diagonal terms** (ϕ_{ra}, ϕ_{ar}) capture cross-state coupling.
- **Stability:** We constrain the spectral radius $\rho(\Phi) < 1$ to ensure stationarity.

3.2 Impulse Occurrence Model

Let $n_j \in \{0, 1\}$ be the binary indicator of an impulse at time j . Occurrence is modeled as a Bernoulli process with a logistic link:

$$n_j \sim \text{Bernoulli}(p_j), \quad \log\left(\frac{p_j}{1-p_j}\right) = \beta + \theta_r x_j^{(r)} + \theta_a x_j^{(a)}$$

Here, θ_a captures whether amplitude-related arousal modulates impulse firing probability.

3.3 Impulse Amplitude Model (Log-Normal)

To enforce physiological positivity ($r_j > 0$), we model the logarithm of the impulse amplitude rather than the amplitude directly. For an observed impulse ($n_j = 1$):

$$\log(r_j) = \gamma_0 + \gamma_r x_j^{(r)} + \gamma_a x_j^{(a)} + v_j, \quad v_j \sim \mathcal{N}(0, \sigma_v^2)$$

This implies that the amplitudes follow a conditional Log-Normal distribution:

$$r_j \mid x_j \sim \text{LogNormal}(\mu_j^{\text{amp}}, \sigma_v^2)$$

where $\mu_j^{\text{amp}} = \gamma_0 + \gamma^\top x_j$. This modification prevents the “negative amplitude” predictions in previous linear-Gaussian formulations.

Interpretability

- $\gamma_r < 0$: higher rate-arousal suppresses amplitude magnitude.
- $\gamma_r > 0$: synchronized rate-amplitude increases.

3.4 Marked Point Process Likelihood

The joint likelihood combines the binary impulse occurrence with the continuous amplitude density (in the log-domain):

$$p(n_j, r_j \mid x_j) = \begin{cases} 1 - p_j, & n_j = 0 \\ p_j \cdot \mathcal{N}(\log r_j; \mu_j^{\text{amp}}, \sigma_v^2), & n_j = 1 \end{cases}$$

4 Inference and Learning

Latent states are inferred using a **Newton-Raphson Extended Kalman Filter** followed by RTS smoothing. The filter linearizes the measurement equation at the mode of the posterior, ensuring robust estimation even for the non-linear Log-Normal amplitude component.

Model parameters $\Theta = \{\Phi, Q, \beta, \theta, \gamma, \sigma_v^2\}$ are learned via Expectation-Maximization (EM):

- **E-step:** Infer posterior distributions over latent states $x_{0:T}$.

- **M-step:** Update parameters using closed-form solutions (for Φ, γ) and numeric optimization (for β, θ). Stability constraints are applied during the M-step to ensure the transition matrix Φ remains stable (eigenvalues within the unit circle) to prevent unbounded state growth.

5 Phase 1: Bernoulli Occurrence with Log-Normal Marks

5.1 Summary

Phase 1 represents the model as described in Sections 3–4. This formulation uses:

- **Bernoulli occurrence model** (Section 3.2) with discrete time bins
- **Log-normal amplitude model** (Section 3.3) ensuring strict positivity
- **Newton-Raphson EKF/RTS** inference (Section 4)

5.2 Advantages

1. **Computational Efficiency:** Discrete-time formulation with straightforward EM updates.
2. **Strict Positivity:** Log-normal marks eliminate negative amplitude predictions.
3. **Stability in Sparse Regions:** When no impulses occur, the amplitude state $x_j^{(a)}$ evolves via VAR(1) dynamics without being constrained by impossible observations.

5.3 Limitations

- **Discrete Time:** Requires binning continuous EDA signals into fixed intervals.
- **Constant Rate Within Bins:** Occurrence probability is constant within each time bin, which may not capture rapid fluctuations in arousal.

6 Phase 2: Inverse Gaussian Point Process with Log-Normal Marks

6.1 Motivation

The Bernoulli occurrence model (Phase 1) assumes discrete time bins and constant occurrence probability within each bin. For EDA, a continuous-time point process provides a more natural representation. We replace the Bernoulli model with an Inverse Gaussian (IG) inter-arrival time distribution while retaining the Log-Normal amplitude model. A key challenge arises because at each spike time t_k , the two available observations depend on states at different times: τ_k depends on $x_{t_{k-1}}$, while r_k depends on x_{t_k} . We resolve this via an **augmented state EKF**, which jointly carries both time points in the filter state and performs a single coherent Newton-Raphson update at each spike, preserving the clean inference structure of Phase 1 without any gradient approximation.

6.2 Model Specification

6.2.1 Inverse Gaussian Inter-Arrival Times

Let $\tau_k = t_k - t_{k-1}$ denote the inter-arrival time between the $(k-1)$ -th and k -th impulse. We model:

$$\tau_k \sim \text{IG}(\mu_{\tau,k}, \lambda), \quad \mu_{\tau,k} = \exp\left(\beta + \theta_r x_{t_{k-1}}^{(r)} + \theta_a x_{t_{k-1}}^{(a)}\right)$$

where the mean inter-arrival time is determined by the latent arousal state at the previous impulse time t_{k-1} .

The IG density is:

$$p(\tau \mid \mu_{\tau}, \lambda) = \sqrt{\frac{\lambda}{2\pi\tau^3}} \exp\left(-\frac{\lambda(\tau - \mu_{\tau})^2}{2\mu_{\tau}^2\tau}\right), \quad \tau > 0$$

Interpretation

- $\theta_r < 0$: higher rate-arousal $\rightarrow$ shorter inter-arrival time $\rightarrow$ higher firing rate
- λ : shape parameter controlling dispersion (larger λ = less variable waiting times)

6.2.2 Log-Normal Amplitude Model (Unchanged from Phase 1)

For an observed impulse at time t_k :

$$\log(r_k) = \gamma_0 + \gamma_r x_{t_k}^{(r)} + \gamma_a x_{t_k}^{(a)} + v_k, \quad v_k \sim \mathcal{N}(0, \sigma_v^2)$$

6.2.3 Continuous-Time State Dynamics

The latent state evolves continuously between observations:

$$dx(t) = A x(t) dt + \Sigma dW(t)$$

where $W(t)$ is a 2-dimensional Brownian motion, A is the drift matrix, and Σ dictates diffusion. The drift matrix is initialized from Phase 1 via the matrix logarithm $A_0 = \frac{1}{\delta} \log \Phi$, where Φ and δ are the transition matrix and bin size learned in Phase 1, providing a principled warm start. For discrete-time filtering on a grid of times $\{t_0, t_0 + \delta, t_0 + 2\delta, \dots\}$:

$$x(t + \delta) = \Phi(\delta) x(t) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, Q_d(\delta))$$

where $\Phi(\delta) = e^{A\delta}$ and the discrete-time covariance is:

$$Q_d(\delta) = \int_0^\delta e^{As} \Sigma \Sigma^\top e^{A^\top s} ds$$

6.3 Complete Likelihood with Survival Function

Let $\mathcal{H}_T = \{(t_1, r_1), (t_2, r_2), \dots, (t_K, r_K)\}$ denote the observed impulse history up to time T . The complete likelihood accounts for observed inter-arrival times and the absence of events after the last impulse:

$$p(\mathcal{H}_T | \Theta) = \prod_{k=1}^K p(\tau_k | x_{t_{k-1}}) p(r_k | x_{t_k}) \cdot S(T - t_K | x_{t_K})$$

where $S(\tau | x)$ is the IG survival function:

$$S(\tau | x) = \Phi \left(-\sqrt{\frac{\lambda}{\tau}} \left(\frac{\tau}{\mu_\tau(x)} - 1 \right) \right) + e^{2\lambda/\mu_\tau(x)} \Phi \left(-\sqrt{\frac{\lambda}{\tau}} \left(\frac{\tau}{\mu_\tau(x)} + 1 \right) \right)$$

with $\Phi(\cdot)$ denoting the standard normal CDF.

6.4 Inference Strategy

6.4.1 E-Step: Augmented State EKF/RTS

Augmented State: At each spike time t_k , define the augmented state:

$$z_k = \begin{bmatrix} x_{t_k} \\ x_{t_{k-1}} \end{bmatrix} \in \mathbb{R}^4$$

Both observations are now explicit functions of a single vector z_k :

- r_k depends on the upper block x_{t_k}
- τ_k depends on the lower block $x_{t_{k-1}}$

This reduces the inference problem to a standard EKF update at a single augmented state, exactly as in Phase 1.

Prediction Step At spike k , propagate the previous filtered state $x_{t_{k-1}}^*$ forward across the interval τ_k :

$$x_{t_k}^{\text{pred}} = \Phi(\tau_k) x_{t_{k-1}}^*, \quad P_{t_k}^{\text{pred}} = \Phi(\tau_k) P_{t_{k-1}}^* \Phi(\tau_k)^\top + Q_d(\tau_k)$$

The augmented predicted mean and covariance are:

$$z_k^{\text{pred}} = \begin{bmatrix} x_{t_k}^{\text{pred}} \\ x_{t_{k-1}}^* \end{bmatrix}, \quad P_z^{\text{pred}} = \begin{bmatrix} P_{t_k}^{\text{pred}} & 0 \\ 0 & P_{t_{k-1}}^* \end{bmatrix}$$

Joint Newton-Raphson Update The MAP estimate of z_k is obtained by maximizing the joint log-posterior:

$$z_k^* = \arg \max_z \left\{ \log p(z \mid z_k^{\text{pred}}, P_z^{\text{pred}}) + \log p(r_k \mid x_{t_k}) + \log p(\tau_k \mid x_{t_{k-1}}) \right\}$$

The gradient and Hessian combine three contributions, each acting on a distinct block of z_k :

- **Prior:**

$$\nabla \log p(z) = -(P_z^{\text{pred}})^{-1}(z - z_k^{\text{pred}}), \quad H_{\text{prior}} = -(P_z^{\text{pred}})^{-1}$$

- **Amplitude** (upper block x_{t_k} only):

$$\nabla_{x_{t_k}} \log p(r_k \mid x_{t_k}) = \frac{\log r_k - \gamma_0 - \gamma^\top x_{t_k}}{\sigma_v^2} \gamma, \quad H_{\text{amp}} = -\frac{\gamma \gamma^\top}{\sigma_v^2}$$

- **Inter-arrival** (lower block $x_{t_{k-1}}$ only), with $\mu_{\tau,k} = \exp(\beta + \theta^\top x_{t_{k-1}})$:

$$\nabla_{x_{t_{k-1}}} \log p(\tau_k \mid x_{t_{k-1}}) = \frac{\lambda(\tau_k - \mu_{\tau,k})}{\mu_{\tau,k}^2} \theta$$

$$H_{\text{IG}} = -\frac{\lambda}{\mu_{\tau,k}^2} \left(1 + \frac{2(\tau_k - \mu_{\tau,k})}{\mu_{\tau,k}} \right) \theta \theta^\top$$

Because amplitude and inter-arrival likelihoods act on separate blocks, the Hessian of the joint log-posterior is block diagonal (up to prior coupling), making each Newton-Raphson iteration efficient. After convergence to z_k^* , the lower block $x_{t_{k-1}}$ is marginalized out and only $x_{t_k}^*$ and its marginal covariance $P_{t_k}^*$ are carried forward to the next interval.

Backward Smoother After the forward filter, a standard RTS backward smoother refines state estimates using future observations, operating on $\{x_{t_k}^*, P_{t_k}^*\}$ identically to Phase 1.

6.4.2 M-Step: Parameter Updates

Continuous-time dynamics (A, Σ) A and Σ are updated in the M-step by maximizing the expected complete-data log-likelihood over state transitions, using smoothed means and cross-covariances from the E-step. Because the augmented filter produces accurate posterior estimates without approximation, A and Σ are identifiable from data and need not be fixed as hyperparameters. Stability is enforced by constraining eigenvalues of A to have strictly negative real parts.

Inter-arrival parameters ($\beta, \theta_r, \theta_a, \lambda$) Maximize the expected complete-data log-likelihood:

$$\mathcal{Q}_{\text{IG}}(\beta, \theta, \lambda) = \mathbb{E}_{p(x_{0:T} \mid \mathcal{H}_T)} \left[\sum_{k=1}^K \log p(\tau_k \mid x_{t_{k-1}}) + \log S(T - t_K \mid x_{t_K}) \right]$$

The expectation is evaluated using the smoothed estimates $\{x_{t_{k-1}}^{\text{smooth}}\}$, which are exact posterior means under the augmented filter. Numerical optimization (e.g., L-BFGS) is used.

Amplitude parameters $(\gamma_0, \gamma_r, \gamma_a, \sigma_v^2)$ Same closed-form ridge regression as Phase 1:

$$\gamma = \left(\sum_{k=1}^K \mathbb{E}[x_{t_k} x_{t_k}^\top] + \lambda_{\text{ridge}} I \right)^{-1} \left(\sum_{k=1}^K \mathbb{E}[x_{t_k}] \log r_k \right)$$